

School Tutor

AI-powered CBSE / NCERT learning, ready for schools and self-learners.

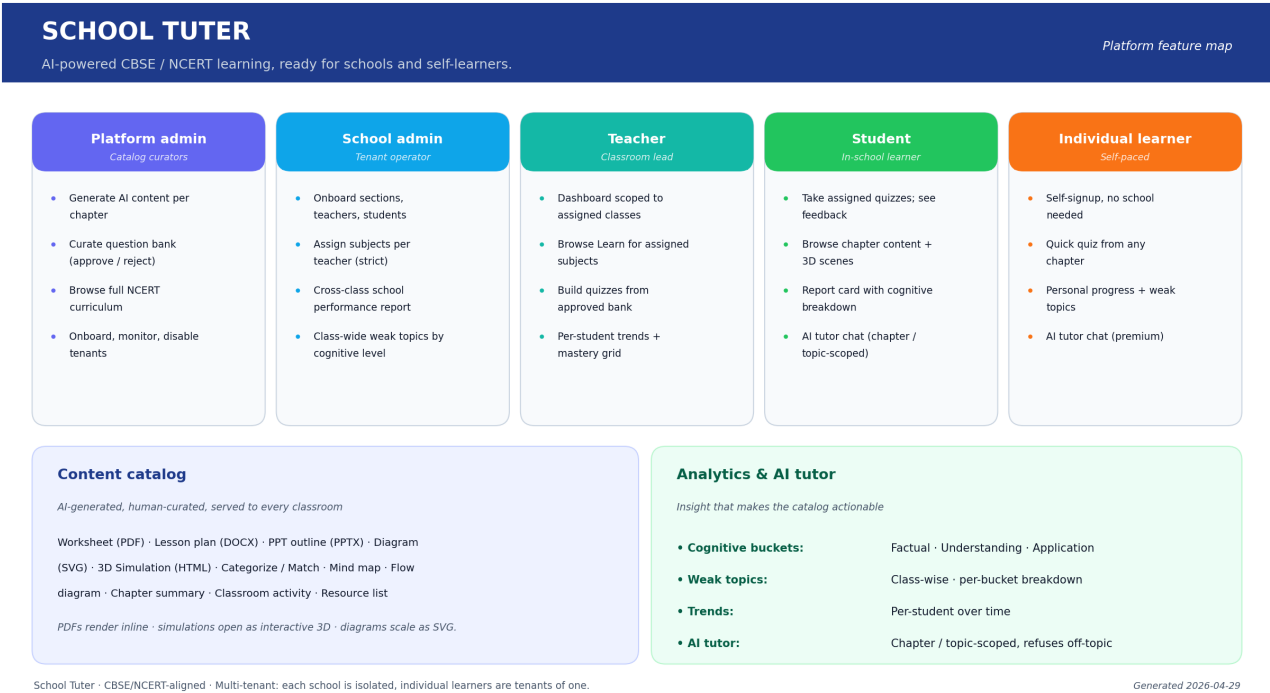
A practical walkthrough of every feature, every role, every content type, and the analytics you can demo to a school or to a parent.

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Executive summary

What School Tutor is, who uses it, and why it matters.

School Tutor is a CBSE/NCERT-aligned platform that turns the official textbook into a complete learning environment — interactive content, practice quizzes, personalised analytics, and an AI tutor — usable by an entire school or by an individual learner. Content is generated by an LLM against the actual chapter text, curated by a small platform team, and then served as a polished, exam-aligned catalog to every classroom. Schools and individuals consume the same library; teachers assemble assessments from a quality-controlled question bank; students get instant feedback, weak-topic insights, and chapter-scoped AI help.



Roles & capabilities

Five distinct roles, each with a tightly scoped surface. The Tutor platform never blurs the line between curation, administration, instruction, and learning.

PLATFORM ADMIN

Catalog curators

- Generate AI content per chapter (worksheets, lesson plans, PPT outlines, diagrams, simulations, mind maps, flow diagrams, classroom activities, resource lists, chapter summaries).
- Curate the question bank — approve, reject, or restore AI-drafted questions before they reach students.
- Browse the full NCERT curriculum: classes, subjects, chapters, learning outcomes, source text.
- Tenant management — list every onboarded school and individual learner, drill into each, disable a tenant in one click (cascades to all its users).
- Toggle the premium AI tutor entitlement per individual learner.

SCHOOL ADMIN

Tenant operator

- Edit school details, brand, contact info.
- Create sections, set class teachers, onboard teachers and students with login credentials.
- Assign teachers to specific subjects (strict scoping — teachers only see what they teach).
- Cross-class school report: leaderboards, average mastery, weak-topic roll-up across every section.
- Class-wise weak topics broken down by Factual / Understanding / Application — pinpoints exactly where to focus.

TEACHER

Classroom lead

- Dashboard with weak topics for the classes they teach, broken down by cognitive level.
- Learn tab scoped to assigned subjects — chapters, topics, summaries, mind maps, simulations.
- Section management — student roster, intervention notes, performance dashboards.
- One-click quiz generation from the approved question bank, with control over difficulty mix, type mix, and cognitive-bucket mix.
- Per-student trend lines and mastery grids.

STUDENT

In-school learner

- Take teacher-assigned quizzes; see immediate score and per-question feedback.
- Browse chapter content — topic summaries, diagrams, flow charts, 3D scenes.
- Personal report card with cognitive-bucket breakdown.
- AI tutor chat scoped to a chapter or single topic — refuses off-topic questions, supports 'Easier please' / 'Go deeper' nudges.
- Practice quick quizzes between formal assessments.

INDIVIDUAL LEARNER

Self-paced

- Self-signup, no school required — same content catalog as a school student.
- Pick any chapter, run a quiz, learn from the topic content.
- Personal dashboard, progress tracking, weak topics.
- AI tutor chat (premium) for one-on-one help.
- Single account per person; platform admin can enable / disable on demand.

Content catalog

Every artefact is generated by an LLM against the actual chapter text, then reviewed and approved by the platform team before it reaches a school.

Type	What it is
Worksheet (PDF)	Printable practice with mixed question types and an answer key.
Lesson plan (DOCX)	Structured teacher script with objectives, recap, lesson body, activities.
PPT outline (PPTX)	Slide-deck outline a teacher can present or polish further.
Diagram (SVG)	Concept maps and labelled diagrams that scale cleanly to projectors.
3D Simulation (HTML)	Three.js scene with arbitrary 3D layouts — atomic models, food webs, plant anatomy, water cycles.
Categorize / Match	Drag-and-drop and match-the-pairs activities.
Mind map	Central-term + radiating branches; great for chapter recap.
Flow diagram	Process flow with labelled transitions — water cycle, life cycles, classification trees.
Chapter summary	Bullet-point recap with key terms and 'why it matters'.
Classroom activity	Hands-on group activity with materials and instructions.
Resource list	Curated supplementary reading and external resources.

Analytics & insights

Every evaluated submission updates a per-student mastery model. The dashboards roll those up for the audience that needs them.

Cognitive buckets

Every question is tagged with one of six Bloom levels which roll up to three coarse buckets: Factual (recall), Understanding (comprehension), and Application (transfer). Mastery is tracked per (student, outcome, bucket), so a teacher can see exactly whether the gap is recall, comprehension, or transfer.

Class-wise weak topics

For any (class, subject) the platform ranks topics by lowest mastery across all accessible students, with the per-bucket breakdown surfaced in line. Available on the teacher Dashboard and the school admin's School Report.

Section leaderboards

Per-section ordered list of student averages and mastery, scoped to subject. Useful for identifying both who needs help and who could lead peer-tutoring.

Per-student trend

Time series of evaluated scores by subject — see whether a student is improving, plateaued, or regressing across assessments.

Mastery heatmap

Per-student grid of (chapter outcome × cognitive bucket) — instantly shows which outcomes need work and at which depth.

AI tutor (premium)

Personal one-on-one help, but rigorously scoped so the model stays on the chapter.

- Premium per-user feature, off by default — toggled on by the platform team.
- Each chat is scoped to either a single chapter or a single topic; the LLM is given just that text and told to refuse off-topic questions.
- Optional difficulty nudges: 'Easier please' simplifies; 'Go deeper' surfaces the precise term plus an extra detail.
- Conversation persists; the student can return to the same chat later.
- Available in English or Hinglish — the tutor matches the student's language.

Multi-tenancy & access control

Every school, every individual, isolated. Platform-admin controls let you onboard, monitor, and revoke access without touching code.

- Every school is a tenant; every individual learner is a 'tenant of one'.
- Cross-tenant data is impossible at the API layer — every query is scoped via the JWT's school_id.
- Platform admin can disable a school in one click; on the next request, every user in that school is rejected at auth.
- Individual learners can be enabled / disabled per user.
- Disabling preserves data — re-enable any time and the user picks up exactly where they left off.

Under the hood

Conventional, boring choices. Easy to hand off, easy to scale.

Layer	Implementation
Backend	FastAPI + SQLAlchemy 2.0 + Alembic on Python 3.12.
Database	PostgreSQL (Supabase). Multi-tenant via row-level school_id scoping.
Frontend	React 19 + Vite 8 + TypeScript + Tailwind CSS v4 + TanStack Query.
LLM	Pluggable: Groq (llama-3.3-70b), OpenAI (gpt-4o-mini), Anthropic (claude-sonnet-4-5).
Hosting	Backend on GCP Cloud Run (scale-to-zero); SPA on Firebase Hosting; artifacts on Cloud Storage.
Auth	JWT (HS256). Per-route role guards. Tenant kill-switch enforced at the auth dependency.

Suggested demo flow

A 7-step walkthrough that hits every major surface in roughly 12 minutes. Use as a script or freestyle.

1. Sign in as platform admin

Show the curriculum browser (Class 6 Science), then the question bank with draft + approved tabs. Click a draft, approve it, watch the badge change.

2. Generate content

Open Generate, pick Class 6 -> Science -> Chapter 1, choose '3D Simulation', click Generate. While it runs, narrate that the LLM uses only the chapter text.

3. Tour the catalog

Open Content library, filter by class + subject + content type, open a worksheet PDF (renders inline) and a 3D simulation (opens in a new tab, interactive).

4. Switch to school admin

Open Manage school -> Sections / Teachers / Students. Open Teachers, click Edit subjects on a teacher, assign a subject, save. Open School report — show class roll-up, leaderboards, weak topics by cognitive bucket.

5. Switch to teacher

Show that the Learn tab and Content library are scoped to assigned subjects only. Open Dashboard — Weak topics card with the Factual / Understanding / Application breakdown.

6. Switch to student

Take a teacher-assigned quiz. After submission, show the report card. Open AI tutor chat from a topic page — ask a question, then ask an off-topic one and watch it politely refuse.

7. Tenants surface

Back as platform admin -> Tenants. Show schools list with student/teacher counts, drill into one, demonstrate the Disable school button (with the confirmation card).